

Operators & Conditional Statements

Learn how JavaScript performs calculations, compares values, and makes decisions — the building blocks of every interactive program.

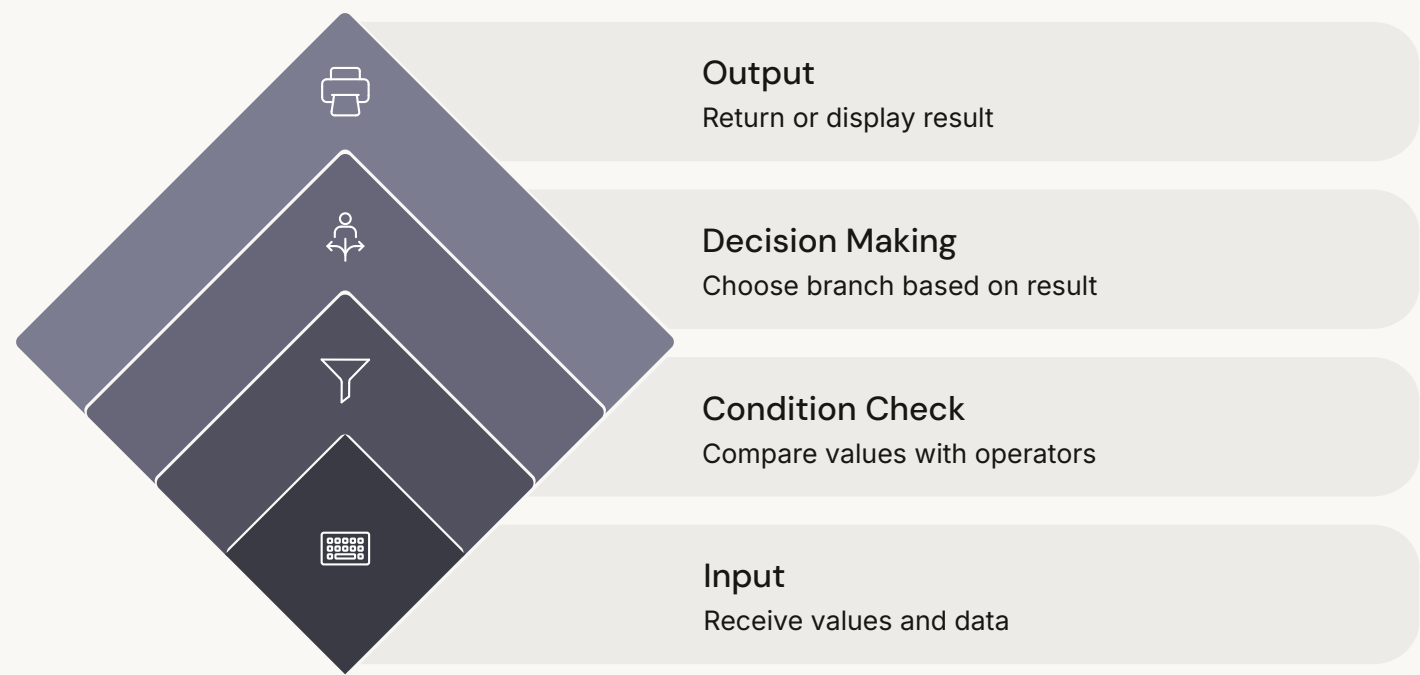
What Will We Learn Today?

Calculations Perform math with operators	Comparisons Compare values and get true/false	Decisions Execute code based on conditions
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Real World Examples

- ATM machines
- Login systems
- Online shopping websites
- Exam result systems
- Voting eligibility systems

Visual Flow



This flow shows how JavaScript processes conditions from receiving input to producing output.

Example

```
let age = 20;  
if(age >= 18){  
  console.log("Eligible to Vote");  
}
```

📌 **Teaching Tip:** Ask students: How does Instagram know whether to show "Login Successful" or "Wrong Password"?

Arithmetic Operators



What are Arithmetic Operators?

Arithmetic operators perform mathematical calculations in JavaScript. They allow you to add, subtract, multiply, divide, find remainders, and calculate powers.

Operators

Operator	Meaning	Example
+	Addition	10 + 5
-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5
%	Modulus	10 % 3
**	Power	2 ** 3

Example

```
let a = 20;
let b = 10;
console.log(a + b);
console.log(a - b);
console.log(a * b);
console.log(a / b);
```

Real World Example

Shopping Cart:

```
let mobilePrice = 15000;
let quantity = 2;
let totalPrice = mobilePrice * quantity;
console.log(totalPrice);
```

📌 **Teaching Tip:** Let students calculate: total marks, mobile bill, and grocery bill using arithmetic operators.

Comparison Operators

Comparison operators compare two values. The result will always be **true** or **false**.

Operators

Operator	Meaning
==	Equal
===	Strict Equal
!=	Not Equal
>	Greater Than
<	Less Than
>=	Greater Than or Equal
<=	Less Than or Equal

Examples

```
console.log(10 > 5);  
// Output: true  
  
console.log(20 < 10);  
// Output: false  
  
let age = 20;  
console.log(age >= 18);  
// Output: true
```

 **Teaching Tip:** Ask students to predict the output before executing each comparison in the console.

Difference Between == and ===

Double Equals (==)

Checks **only value** — ignores data type.

```
console.log(10 == "10");  
// Output: true
```

Triple Equals (===)

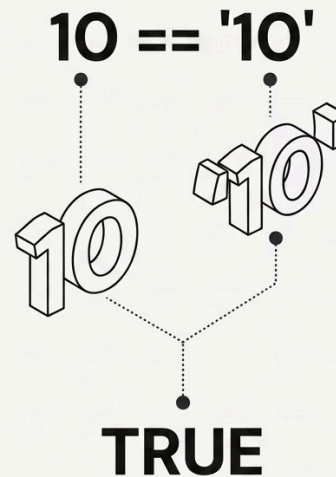
Checks both **value** and **data type**.

```
console.log(10 === "10");  
// Output: false
```

✅ **Industry Best Practice:** Always use `===`. Most companies prefer strict comparison to avoid unexpected type coercion bugs.

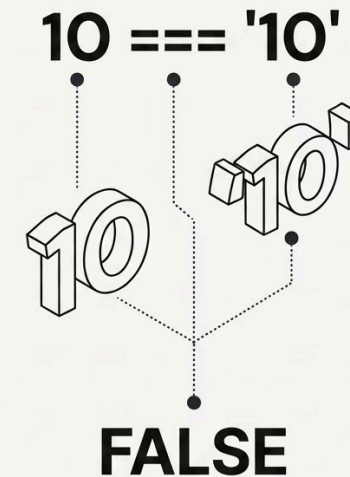
Visual Understanding

Double Equals



Checks **VALUE** only,
returns **TRUE**.

Triple Equals



Checks **VALUE** and
TYPE, returns **FALSE**.

📌 **Teaching Tip:** This is one of the most common JavaScript interview questions. Make sure students understand the difference clearly.

Logical Operators

Logical operators combine multiple conditions to form more complex decision-making logic.

Types

Operator	Meaning
&&	AND
	OR
!	NOT

AND Operator (&&)

Both conditions must be **true**.

```
let age = 20;
console.log(age > 18 && age < 30);
// Output: true
```

OR Operator (||)

At least **one** condition must be true.

```
console.log(age < 18 || age > 60);
// Output: false
```

📌 **Teaching Tip:** Use college admission examples — e.g., "Must have 75% AND pass the interview" or "Have a scholarship OR pay fees."

Conditional Statements: if

What is the if Statement?

The if statement executes code **only when the condition is true**. If the condition is false, nothing happens.

Syntax

```
if(condition){  
  // code executes when true  
}
```

Example

```
let age = 20;  
if(age >= 18){  
  console.log("Eligible to Vote");  
}
```

Real World Usage

- Voting eligibility
- Login validation
- Exam qualification

📌 **Teaching Tip:** Create multiple age examples (15, 18, 25) to show how the condition changes the output.



if else Statement

Used when there are **two possible outcomes** — one path for true, another for false.

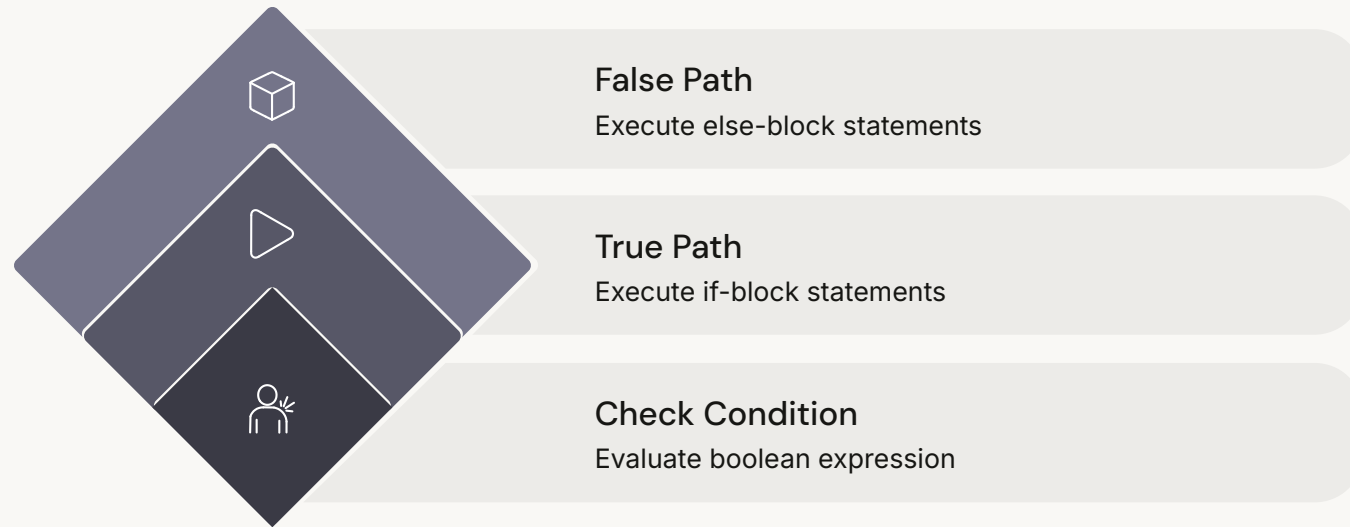
Example

```
let age = 16;  
if (age >= 18){  
  console.log("Eligible");  
}  
else{  
  console.log("Not Eligible");  
}
```

Real World Example — ATM Withdrawal

```
if (balance >= amount){  
  console.log("Transaction Successful");  
}  
else{  
  console.log("Insufficient Balance");  
}
```

Flow



The if else statement ensures one of two code blocks always executes based on the condition result.

📌 **Teaching Tip:** Relate this to real ATM transactions — students instantly understand insufficient balance scenarios.

else if Statement

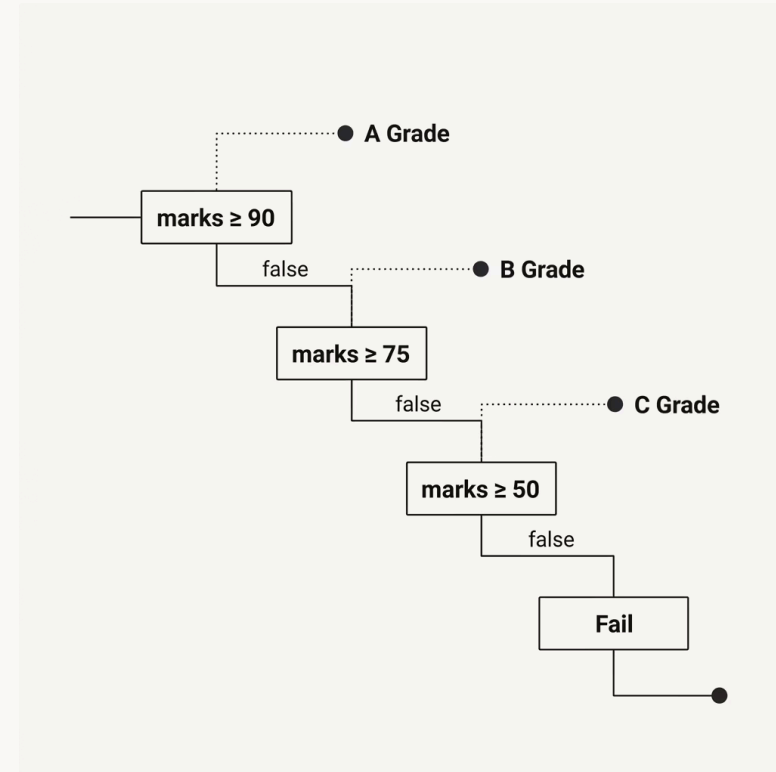
Used when **multiple conditions** exist. JavaScript checks each condition in order until one is true.

Example — Grade System

```
let marks = 85;  
if(marks >= 90){  
  console.log("A Grade");  
}  
else if(marks >= 75){  
  console.log("B Grade");  
}  
else if(marks >= 50){  
  console.log("C Grade");  
}  
else{  
  console.log("Fail");  
}
```

Real World Usage

- Grade systems
- Tax calculations
- Employee bonuses



📌 **Teaching Tip:** Build a Grade Calculator live in class — let students input different marks and observe the output change.

switch Statement

What is switch?

The `switch` statement is used when checking **fixed values**. It's a cleaner alternative to many `else if` conditions.

Example — Day Finder

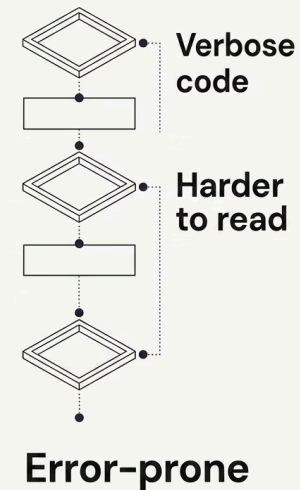
```
let day = 3;
switch(day){
  case 1:
    console.log("Monday");
    break;
  case 2:
    console.log("Tuesday");
    break;
  case 3:
    console.log("Wednesday");
    break;
  default:
    console.log("Invalid Day");
}
```

Real World Usage

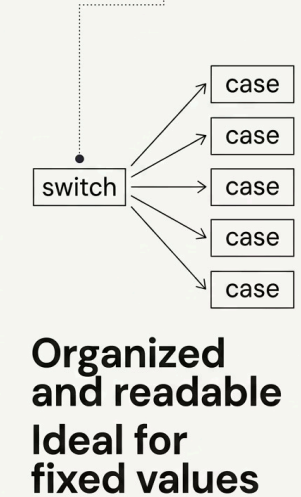
- Language selection menus
- App menu options
- User role routing
- Day finder apps

📌 **Teaching Tip:** Have students create a Day Finder Program that maps numbers 1–7 to day names.

Many else if conditions



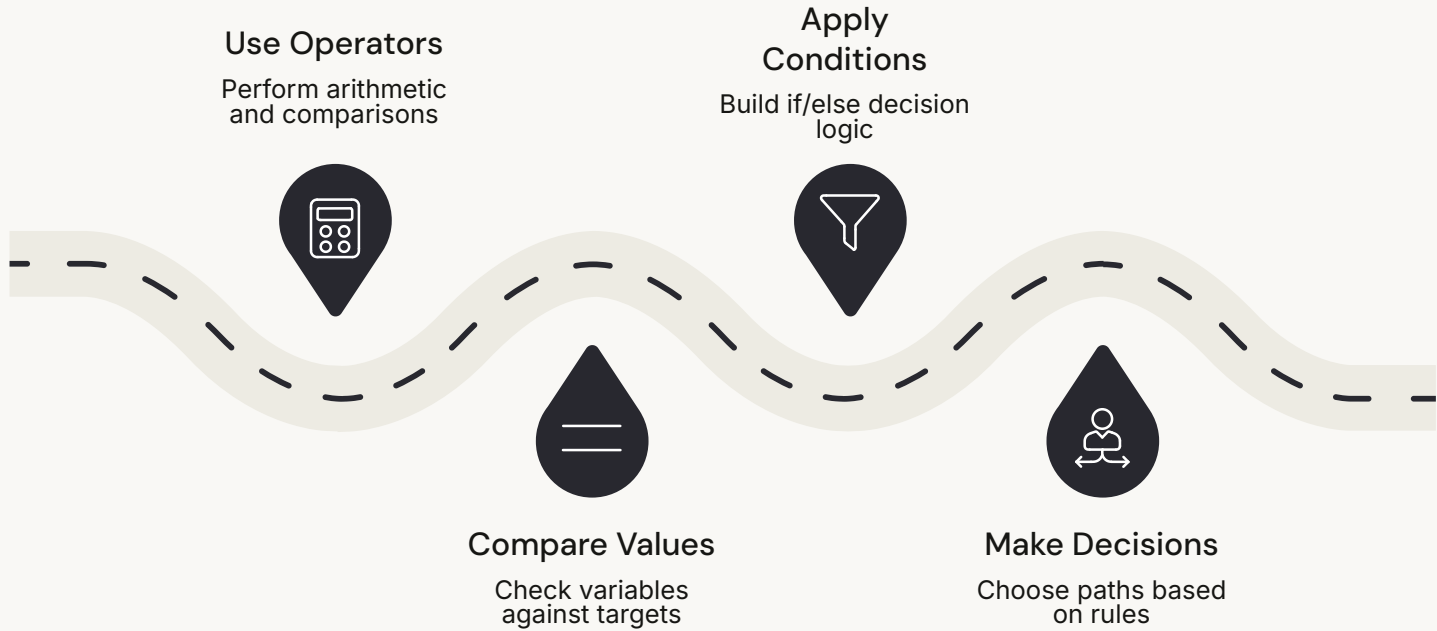
Clean switch statement



Mini Tasks, Homework & Outcomes

1	2	3
Mini Task 1 — Grade Calculator <pre>let marks = 88; // Expected Output: B Grade</pre>	Mini Task 2 — Voting Eligibility <pre>let age = 20; // Expected Output: Eligible to Vote</pre>	Homework — ATM Balance Checker <pre>let balance = 5000; let withdrawAmount = 2000; // Store balance & withdrawal amount // Check balance condition // Show success or failure message</pre>

Final Outcome — After Day 2, Students Will:



Students will have mastered the fundamentals of JavaScript logic — from basic arithmetic to full decision-making programs used in real-world applications.